

Homework — 1.2.3 Software Development

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 1.2.3 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. State, in the correct order, the five main stages of the **waterfall** lifecycle. [2] (AO1)

2. A company is building software for a client whose requirements are **likely to change frequently** during the project. Explain why an **agile** methodology is more suitable than the **waterfall** model for this project. [3] (AO2)

3. **Extreme Programming (XP)** is an Agile approach. Describe **two** practices that are characteristic of XP, and state **one** type of project for which XP is well suited. [3] (AO1 / AO2)

4. A program accepts a percentage mark that must be between 0 and 100 inclusive. Give one example of **boundary**, **normal (valid)** and **erroneous** test data for this input, and explain why boundary testing is especially important here. [4] (AO2)

5. Describe how a system is developed using **Rapid Application Development (RAD)**, and state **one** drawback of this methodology. [2] (AO1)

Part B — Retrieval: keep it fresh (~6 marks)

7. (from 1.2.1) A **washing machine** and a **server cluster** use different types of operating system. Name the most suitable OS type for each and justify your choice. [2] (AO2)

8. (from 1.1.3) Explain the difference between **RAM and **ROM** in terms of volatility and typical use. [2]**
(AO1)
